## Design vs. Sandbox

The Part 1 AWS architecture represents the enterprise solution for a production environment. It is designed to support growth from three to twelve plants, protect export-controlled data, enforce least-privilege access, and provide a data and AI platform. Part 3 is a scaled-down implementation completed in the AWS Academy Learner Lab. Its purpose is to demonstrate that the major components of the architecture work together, rather than reproduce every enterprise feature.

### What Part 3 Simplifies and Why

- **Security and permissions:** IAM roles, multi-account separation, least-privilege policies, and detailed Lake Formation permissions are used in the enterprise design. The sandbox may use the provided LabRole to access AWS services because the learning environment has predefined permissions and restrictions. This makes implementation easier, but it does not demonstrate the full security controls required for production.
- **AI capabilities:** Part 1 includes Amazon Bedrock, AgentCore, knowledge bases, guardrails, and multiple specialized AI agents. Part 3 focuses on implementing the core data pipeline and selected analytics capabilities. A Bedrock agent environment is not needed to show the ingestion that is required.
- **Data volumes:** The architecture must be able to handle approximately 2,400 machines across twelve plants and large-scale telemetry streams. The sandbox uses the instructor-provided data pack, which contains a representative sample of 24 instrumented machines across three plants. This means the team can test ingestion and processing without enterprise-scale workloads.
- **Networking and disaster recovery:** The production design includes multi-account boundaries, cross-Region recovery, and plant connectivity. The sandbox focuses on getting the required services running within the available Learner Lab environment rather than implementing the full enterprise network and DR architecture.
- **Production automation and governance:** The design includes monitoring, data governance, automated recovery, compliance controls, and deployment pipelines. The sandbox has selected automation, monitoring, and cost-management practices within the time and resource limits of the course.

Part 3 is supposed to support the technical concepts and establish a working version of the solution, not to create a production-ready platform. The AWS Academy has limited permissions, available services, time, and budget. Implementing every enterprise feature would add intricacy and cost without changing the learning. The sandbox should still show the Plant-to-P&L workflow: ingesting data, storing the data in S3, transforming it with Glue, querying it through the warehouse, applying analytics or ML, and monitoring the pipeline.